

PHYSICS QUEST

A Gamified Learning Series

FORCE

and the Laws of Motion

Module 3 — Grade 9 Physics Companion



This is a fictional, original teaching resource
created for classroom gamification & hackathon demo purposes.

Physics Quest • Unit 3 • Sample Edition

Module 3

Force and the Laws of Motion

In this module, you will explore one of the most fundamental ideas in physics: **force**. Forces are all around us — they let us walk, throw a ball, drive a car, and even keep planets in orbit. By the end of this module you will be able to identify different types of forces, apply Newton's three laws of motion, draw free-body diagrams, and calculate net force in simple systems.

What You Will Learn

- Define force and identify its effects on objects
- Distinguish between contact and non-contact forces
- State and apply Newton's First, Second, and Third Laws of Motion
- Draw and interpret free-body diagrams
- Explain friction and its role in everyday motion
- Calculate net force and determine equilibrium

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3.1 What Is a Force?

A **force** is a push or a pull acting on an object, resulting from its interaction with another object. Forces can change an object's speed, direction, or shape. Force is a **vector** quantity, meaning it has both magnitude (size) and direction. The SI unit of force is the **newton (N)**, named after Sir Isaac Newton.

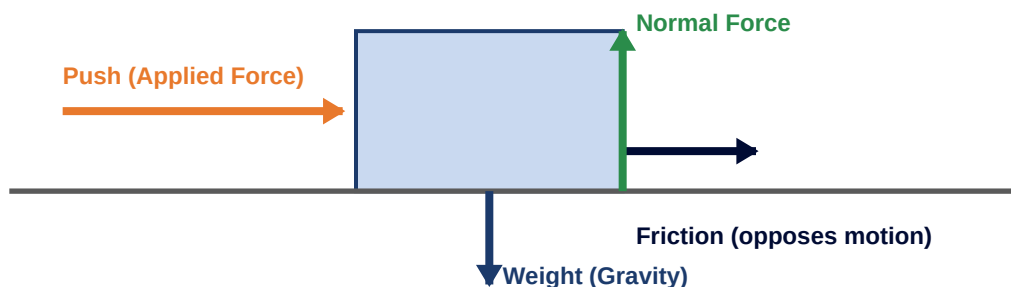


Figure 3.1 — Several forces acting on a box resting on the ground.

Whenever you push a door open, kick a ball, or feel gravity pull you toward the ground, you are experiencing a force. Even when an object looks still, forces may be acting on it — they are simply balanced.

Key Idea

A force cannot exist without an interaction between two objects. Every force has a source and a receiver — something doing the pushing/pulling, and something being pushed or pulled.

3.2 Types of Forces

Forces are generally grouped into two categories: **contact forces**, which require physical touching between objects, and **non-contact (field) forces**, which act across a distance.

Category	Force	Example
Contact	Applied Force	Pushing a shopping cart
Contact	Friction	A sled slowing down on snow
Contact	Normal Force	A book resting on a table
Contact	Tension	A rope pulling a sled
Non-contact	Gravitational Force	Earth pulling you downward
Non-contact	Magnetic Force	A magnet attracting a paperclip
Non-contact	Electrostatic Force	Static hair attracted to a balloon

Try This

Look around you right now. Name one contact force and one non-contact force acting on an object near you. (Hint: your chair and gravity are a good place to start!)

3.3 Newton's First Law of Motion — Inertia

Newton's First Law states that an object at rest stays at rest, and an object in motion stays in motion at constant velocity, unless acted upon by a net external force. This property of matter is called **inertia** — the tendency to resist changes in motion.



Figure 3.2 — Objects resist changes to their state of motion.

This is why passengers lurch forward when a car suddenly stops — their bodies were in motion and tend to continue moving even though the car has stopped.

Real-World Connection

Seatbelts exist because of inertia! In a collision, your body wants to keep moving forward even though the car has stopped. The seatbelt provides the force needed to stop you safely.

3.4 Newton's Second Law of Motion — $F = ma$

Newton's Second Law describes the relationship between force, mass, and acceleration. It states that the acceleration of an object is directly proportional to the net force acting on it, and inversely proportional to its mass.

$$F = m \times a$$

Where **F** is net force (in newtons, N), **m** is mass (in kilograms, kg), and **a** is acceleration (in meters per second squared, m/s^2).

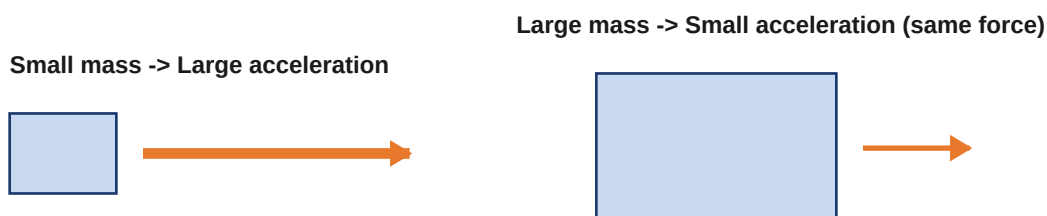


Figure 3.3 — The same force produces different accelerations depending on mass.

Worked Example

A 10 kg cart is pushed with a net force of 20 N. What is its acceleration?

$$a = F \div m = 20 \text{ N} \div 10 \text{ kg} = \mathbf{2 \text{ m/s}^2}$$

3.5 Newton's Third Law of Motion — Action & Reaction

Newton's Third Law states that for every action force, there is an equal and opposite reaction force. These two forces act on *different* objects and occur simultaneously.

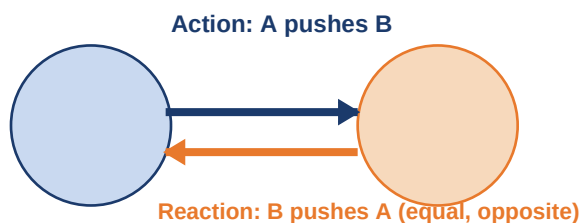


Figure 3.4 — Object A exerts a force on Object B; Object B pushes back equally.

A classic example is a swimmer pushing water backward with their arms (action); the water pushes the swimmer forward with equal force (reaction). Rockets also use this principle, expelling gas downward to be propelled upward.

Quick Check

If a book pushes down on a table with 5 N of force, how much force does the table push back up on the book? (Answer at the end of this chapter.)

3.6 Free-Body Diagrams

A **free-body diagram (FBD)** is a simplified drawing that shows all the forces acting on a single object, represented as arrows. The length of each arrow roughly represents the size of the force, and the direction shows which way the force acts.

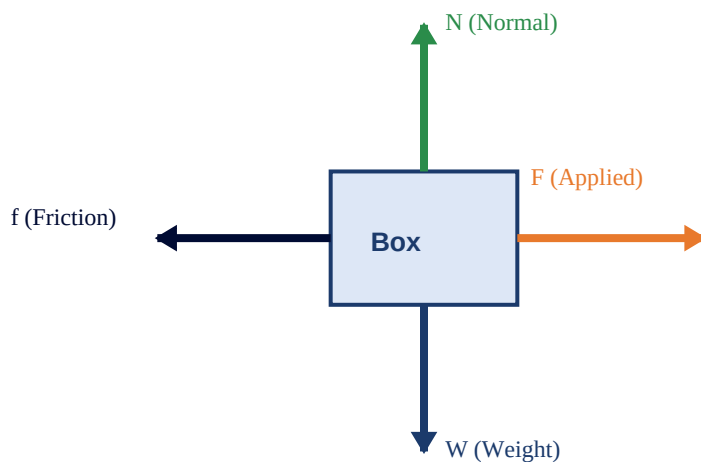


Figure 3.5 — A free-body diagram of a box being pushed across a rough floor.

Steps to draw a free-body diagram:

- 1 Represent the object as a simple shape (a box or dot).
- 2 Identify every force acting on the object.
- 3 Draw each force as an arrow starting from the object, pointing in its direction of action.
- 4 Label each arrow with the force name and, if known, its magnitude.

3.7 Friction

Friction is a contact force that opposes the relative motion (or attempted motion) between two surfaces in contact. It arises from microscopic irregularities between surfaces.



Figure 3.6 — Friction acts opposite to the direction of applied motion.

Type	Description
Static Friction	Prevents an object from starting to move
Kinetic (Sliding) Friction	Acts on an object already in motion
Rolling Friction	Acts on objects that roll, e.g. wheels or balls

Did You Know?

Without friction, you would not be able to walk, cars could not grip the road, and even writing with a pencil would be impossible!

3.8 Net Force and Equilibrium

The **net force** is the overall force acting on an object after combining all individual forces, taking direction into account. When the net force on an object is zero, the object is said to be in **equilibrium** — it will remain at rest or continue moving at constant velocity.

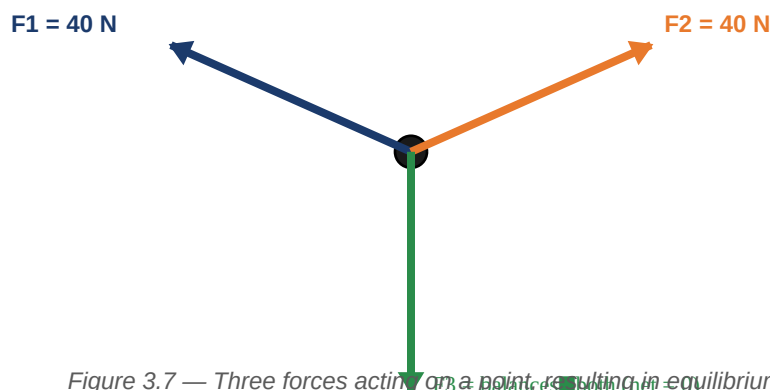


Figure 3.7 — Three forces acting on a point, resulting in equilibrium (net force = 0).

To find net force along a straight line, add forces acting in the same direction and subtract forces acting in the opposite direction.

Worked Example

Two students push a box: one with 30 N to the right, another with 12 N to the left. What is the net force?

Net Force = $30\text{ N} - 12\text{ N} = 18\text{ N to the right}$

3.9 Worked Examples

Example 1: Calculating Acceleration

A 5 kg skateboard is pushed with a net force of 15 N. Find its acceleration.

$$a = F \div m = 15 \div 5 = \mathbf{3 \text{ m/s}^2}$$

Example 2: Calculating Force

A 1200 kg car accelerates at 2 m/s². What net force is required?

$$F = m \times a = 1200 \times 2 = \mathbf{2400 \text{ N}}$$

Example 3: Identifying Balanced Forces

A picture frame hangs motionless on a wall. The string tension pulls up with 8 N. What is the weight of the frame?

Since the frame is in equilibrium, weight = tension = **8 N**

3.10 Chapter Summary

- A force is a push or pull that can change an object's motion or shape; measured in newtons (N).
- Forces are contact (friction, tension, normal, applied) or non-contact (gravity, magnetic, electrostatic).
- Newton's First Law: objects resist changes in motion (inertia).
- Newton's Second Law: $F = m \times a$ — force equals mass times acceleration.
- Newton's Third Law: every action has an equal and opposite reaction.
- Free-body diagrams help visualize all forces acting on an object.
- Friction opposes relative motion between surfaces.
- When net force = 0, an object is in equilibrium.

Key Vocabulary

Force • Newton (N) • Inertia • Mass • Acceleration • Friction • Tension • Normal Force • Net Force • Equilibrium • Free-Body Diagram

3.11 Quest Challenge — Practice Questions

Complete these challenges to earn Force Points! Each correct answer unlocks progress in your learning quest.

Q1. (5 pts) Which of the following is a non-contact force?

- A) Friction
- B) Tension
- C) Gravity
- D) Normal Force

Q2. (10 pts) A 4 kg object experiences a net force of 12 N. Calculate its acceleration.

- A) 3 m/s²
- B) 4 m/s²
- C) 8 m/s²
- D) 48 m/s²

Q3. (5 pts) Which law explains why a passenger jerks forward when a bus stops suddenly?

- A) First Law
- B) Second Law
- C) Third Law
- D) Law of Gravitation

Q4. (10 pts) Two forces act on a box: 25 N right and 10 N left. What is the net force?

- A) 35 N left
- B) 15 N right
- C) 15 N left
- D) 35 N right

Q5. (5 pts) What force always opposes the relative motion between two surfaces?

- A) Gravity
- B) Tension
- C) Friction
- D) Normal Force

Q6. (10 pts) A rocket expels gas downward and accelerates upward. Which law explains this?

- A) First Law
- B) Second Law
- C) Third Law
- D) None of the above

Q7. (5 pts) An object is in equilibrium when:

- A) It is moving very fast
- B) Net force = 0

- C) It has no mass
- D) It is touching another object

Bonus Challenge (15 pts)

Draw your own free-body diagram for a book sliding across a rough table after being given a push. Label all four forces acting on it.

Answer Key

Question	Answer	Explanation
Q1	C — Gravity	Gravity acts at a distance; no contact needed.
Q2	A — 3 m/s^2	$a = F/m = 12/4 = 3 \text{ m/s}^2$
Q3	A — First Law	Inertia keeps the passenger's body moving forward.
Q4	B — 15 N right	$25 \text{ N} - 10 \text{ N} = 15 \text{ N}$ in the direction of the larger force.
Q5	C — Friction	Friction always opposes relative sliding motion.
Q6	C — Third Law	Gas pushed down (action); rocket pushed up (reaction).
Q7	B — Net force = 0	Equilibrium means all forces balance out.

Quick Check answer (from Section 3.5): The table pushes back on the book with exactly 5 N, equal and opposite to the book's force on the table.

End of Module 3. Continue to Module 4: Energy and Work in the Physics Quest series.